A Comparative Study of Deep Learning Architectures for Facial Emotion Recognition Using the FERPlus Dataset

Nada Ashraf, Aly Zaki, Omar Bayoumi, and Ahmed Waleed

Faculty of Computing and Information Systems
Artificial Intelligence Track
Egypt University of Informatics
Cairo, Egypt

Supervised by: Dr. Mona Soliman

Abstract :

Facial emotion recognition is a fundamental problem in computer vision with applications in human–computer interaction, affective computing, and intelligent systems. Despite recent advances, accurate emotion classification remains challenging due to subtle facial expressions, class imbalance, and noisy labels. In this work, we present a comparative study of eight deep learning architectures for facial emotion recognition using the FERPlus dataset. The evaluated models span baseline convolutional neural networks trained from scratch and transfer learning architectures leveraging pretrained ImageNet weights, including ResNet50, MobileNetV2, EfficientNet, and DenseNet. All models are trained and evaluated under a unified experimental protocol to ensure fair comparison. Performance is assessed using accuracy, precision, recall, F1-score, and confusion matrix analysis. Our results demonstrate that transfer learning models generally outperform baseline architectures, while lightweight models offer competitive performance with reduced computational cost. The study highlights key trade-offs between model complexity, generalization ability, and efficiency for facial emotion recognition tasks.

Index Terms: Facial Emotion Recognition, Deep Learning, FERPlus, Convolutional Neural Networks, Transfer Learning.

1 Introduction

Facial emotion recognition (FER) is a fundamental task in computer vision that focuses on identifying human emotional states from facial expressions. It plays an important role in applications such as human computer interaction, affective computing, intelligent tutoring systems, mental health assessment, and social robotics. Despite significant progress, reliable emotion recognition remains challenging due to subtle facial muscle movements, inter-class similarity between emotions, variations in lighting and pose, occlusions, and ambiguity in human expressions.

Deep learning, particularly convolutional neural networks (CNNs), has become the dominant paradigm for image-based recognition tasks due to its ability to learn hierarchical feature representations directly from data. In facial emotion recognition, CNN-based approaches have shown clear advantages over traditional handcrafted-feature methods. However, model performance is strongly influenced by architectural design, training strategy, data preprocessing, and dataset characteristics. As a result, systematic evaluation across different architectures is essential to understand their strengths and limitations.

The FERPlus dataset is a widely adopted benchmark for facial emotion recognition and serves as the foundation of this study. FERPlus extends the original FER2013 dataset by providing improved label quality through crowdsourced annotations, resulting in more reliable emotion labels. The dataset consists of low-resolution facial images categorized into multiple emotion classes. Despite its popularity, FERPlus presents several challenges, including class imbalance, noisy samples, and visually similar emotion categories such as fear and surprise or anger and disgust. These characteristics make FERPlus a suitable and challenging benchmark for evaluating the robustness and generalization ability of deep learning models.

This work presents a FERPlus-based comparative study of deep learning architectures for facial emotion recognition. We evaluate eight models belonging to two main categories: baseline convolutional neural networks trained from scratch and transfer learning models leveraging pretrained ImageNet weights. The baseline models provide insight into the effectiveness of shallow and moderately deep architectures when trained solely on FERPlus, while the transfer learning models assess the benefit of feature reuse from large-scale datasets. The selected architectures include both lightweight and deep models, allowing analysis of trade-offs between accuracy, generalization, and computational efficiency.

To ensure fairness and reproducibility, all models are evaluated using a unified experimental protocol and common evaluation metrics, including accuracy, precision, recall, and F1-score. In addition to aggregate performance metrics, confusion matrices and misclassification analysis are used to examine class-wise behavior and identify common error patterns specific to FERPlus. Through this analysis, the study aims to provide practical insights into selecting suitable deep learning architectures for FERPlus-based facial emotion recognition tasks.

The remainder of this paper is organized as follows. Section 2 reviews relevant background related to facial emotion recognition and deep learning. Section 3 describes the FERPlus dataset, preprocessing steps, and experimental setup. Section 4 presents the evaluated model architectures and training configurations. Section 5 reports the experimental results and comparative analysis. Section 6 discusses the findings and limitations of the study, followed by conclusions and future work.

2 Background

2.1 Facial Emotion Recognition

Facial emotion recognition (FER) is a subfield of computer vision and affective computing that aims to automatically identify human emotional states from facial expressions. Early approaches relied on handcrafted features such as facial landmarks, geometric descriptors, and texture-based representations combined with traditional machine learning classifiers. While these methods achieved reasonable performance, they were sensitive to variations in illumination, pose, occlusion, and subject-specific facial characteristics.

The introduction of deep learning significantly advanced FER by enabling automatic feature learning directly from image data. Convolutional neural networks (CNNs) have become the dominant architecture due to their ability to capture spatial hierarchies and local facial patterns. CNN-based models have demonstrated improved robustness and generalization compared to traditional methods. Nevertheless, FER remains challenging because emotional expressions can be subtle, ambiguous, and visually similar across different emotion categories.

2.2 FERPlus Dataset

The FERPlus dataset is a widely used benchmark for facial emotion recognition and serves as an extension of the FER2013 dataset. FER2013 contains grayscale facial images labeled into emotion categories but suffers from noisy and inconsistent annotations. FERPlus addresses this limitation by introducing a crowdsourced labeling process, where each image is annotated by multiple human annotators, leading to improved label reliability.

The dataset consists of low-resolution facial images categorized into multiple emotion classes, including happiness, sadness, anger, fear, surprise, disgust, and neutral expressions. Despite improved annotation quality, FERPlus remains challenging due to class imbalance, ambiguous samples, and visually similar emotion pairs such as fear and surprise or anger and disgust. These characteristics make FERPlus a suitable benchmark for evaluating the robustness of deep learning-based FER models.

2.3 Deep Learning Architectures for FER

Deep learning architectures for FER range from shallow baseline CNNs trained from scratch to deeper networks incorporating advanced design principles. Baseline CNNs typically consist of a small number of convolutional and pooling layers followed by fully connected layers. These models serve as useful benchmarks and help analyze the impact of network depth when trained exclusively on FERPlus.

More advanced architectures introduce innovations such as residual connections and inception-style modules. Residual networks facilitate training deeper models by improving gradient flow, while inception-based architectures enable multi-scale feature extraction. These designs have proven effective in general image recognition tasks and are increasingly adopted in FER applications.

2.4 Transfer Learning in Facial Emotion Recognition

Transfer learning is widely used in FER to address limitations related to dataset size and variability. In this approach, models pretrained on large-scale datasets such as ImageNet are fine-tuned on FER datasets. Pretrained models benefit from rich feature representations that improve convergence speed and generalization performance.

Popular transfer learning architectures include ResNet, MobileNet, EfficientNet, and DenseNet. These models differ in depth, parameter count, and computational complexity, enabling exploration of trade-offs between accuracy and efficiency. While transfer learning often improves FER performance, deeper models may introduce higher computational cost and risk of overfitting if not carefully configured.

2.5 Motivation for Comparative Evaluation

Given the diversity of available architectures and training strategies, systematic comparative evaluation is essential to understand their relative strengths and limitations in facial emotion recognition. Architectural design choices can significantly affect accuracy, robustness, and computational efficiency. Conducting a comparative study on a consistent dataset such as FERPlus enables fair and meaningful evaluation.

By evaluating both baseline and transfer learning models under a unified experimental protocol, this work aims to provide insights into how architectural choices influence performance on FERPlus. Such analysis is particularly valuable in academic and applied contexts where selecting an appropriate model requires balancing performance and practical constraints.

3 Related Work

Recent advances in facial emotion recognition have been largely driven by deep learning, particularly convolutional neural networks (CNNs), which enable end-to-end learning of discriminative facial features directly from raw image data. Early CNN-based approaches demonstrated substantial improvements over traditional handcrafted feature methods by capturing spatial hierarchies and local facial patterns associated with emotional expressions (Krizhevsky, Sutskever, and Hinton). To address limitations in network depth and training stability, more advanced architectures such as residual networks and inception-based models were introduced, improving gradient flow and enabling multi-scale feature extraction, which proved beneficial for recognizing subtle and visually similar emotions (He et al.). The FERPlus dataset, introduced as an enhancement to the FER2013 dataset, further advanced FER research by providing improved label quality through crowdsourced annotations, leading to more reliable supervision and improved performance of deep learning models (Barsoum et al.). Following its release, FERPlus became a common benchmark for evaluating CNN-based emotion recognition systems, although challenges such as class imbalance, ambiguous expressions, and confusion between similar emotions such as fear and surprise remain prevalent. To mitigate data limitations and improve generalization, transfer learning has been widely adopted in facial emotion recognition, where models pretrained on large-scale datasets such as ImageNet are fine-tuned on FERPlus, often yielding superior performance compared to training from scratch (Razavian et al.). Popular architectures including ResNet, MobileNet, DenseNet, and EfficientNet have been applied to FER tasks, revealing important trade-offs between model complexity, accuracy, and computational efficiency. Despite these advancements, open challenges persist, including sensitivity to dataset bias, limited generalization to real-world conditions, and inconsistent performance across emotion classes, highlighting the need for systematic comparative studies that evaluate multiple architectures under unified experimental settings.

4 Dataset

This study is based on the FERPlus dataset, a widely used benchmark for facial emotion recognition. FERPlus is an extension of the original FER2013 dataset and was introduced to address limitations related to noisy and inconsistent labels. Unlike FER2013, FERPlus employs a crowdsourced labeling process in which each facial image is annotated by multiple human annotators, resulting in more reliable emotion labels.

The dataset consists of grayscale facial images captured in unconstrained conditions, including variations in pose, illumination, facial expression intensity, and occlusions. Images are provided at low resolution and are categorized into multiple emotion classes, including happiness, sadness, anger, fear, surprise, disgust, and neutral expressions. These characteristics make FERPlus a challenging dataset that closely reflects real-world facial emotion recognition scenarios.

Figure 1 shows representative samples from the FERPlus dataset, highlighting the diversity of facial expressions and image conditions.

It is important to note that while all models are trained and evaluated on the same FERPlus dataset and data splits, preprocessing strategies may differ across architectures. Certain models require specific input resolutions, normalization schemes, or data augmentation techniques to accommodate architectural constraints or improve performance. As a result, preprocessing steps are described within the corresponding model sections rather than as part of a global preprocessing pipeline.

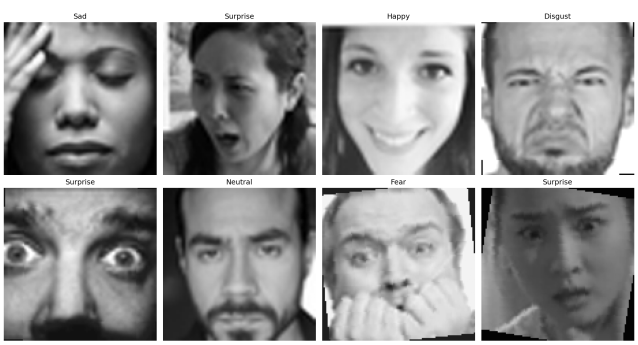

Figure 1. Sample images from the FERPlus dataset illustrating variations in facial expressions, pose, illumination, and emotion categories.

5 Methodology (Models from Scratch)

Due to differences in architectural requirements and individual implementation choices, preprocessing and training configurations are described separately for each model. All models are evaluated on the same FERPlus dataset and test set to ensure comparability of results.

5.1 GAP-Regularized Stronger CNN (GAP-SCNN) Bayoumi

5.1.1 Data Preprocessing FERPlus grayscale images were converted to three-channel RGB format to ensure compatibility with standard CNN input conventions. Image resizing, normalization, and batching were applied uniformly, while data augmentation was applied only to the training set.

Table 1
Preprocessing pipeline for GAP-SCNN.

Step	**Details**	**Rationale**
Image resizing	$224 \times 224 \times 3$	Standard input size for modern CNNs
Pixel normalization	$[0, 255] \rightarrow [0, 1]$	Numerical stability and faster convergence
Color format	RGB (3-channel)	Converted from grayscale FERPlus images
Batch size	64	Efficient GPU utilization

Table 2
Training-set data augmentation strategy for GAP-SCNN.

Technique	**Parameters**	**Purpose**
Random horizontal flip	$p = 0.5$	Account for facial symmetry
Random rotation	$\pm 10°$	Handle head tilts
Random zoom	$\pm 10\%$	Simulate distance variation

Validation and test sets were not augmented to ensure fair evaluation.

5.1.2 Model Architecture The GAP-SCNN consists of four convolutional blocks with increasing filter depths (32, 64, 128, 256). Each block contains a 3×3 convolution, Batch Normalization, ReLU activation, and 2×2 MaxPooling.

A **Global Average Pooling (GAP)** layer replaces the traditional flatten operation, significantly reducing parameter count and mitigating overfitting. The classifier head includes a fully connected layer with 512 units followed by Batch Normalization, ReLU activation, Dropout, and L2 regularization. The output layer uses an 8-unit softmax activation corresponding to FERPlus emotion classes.

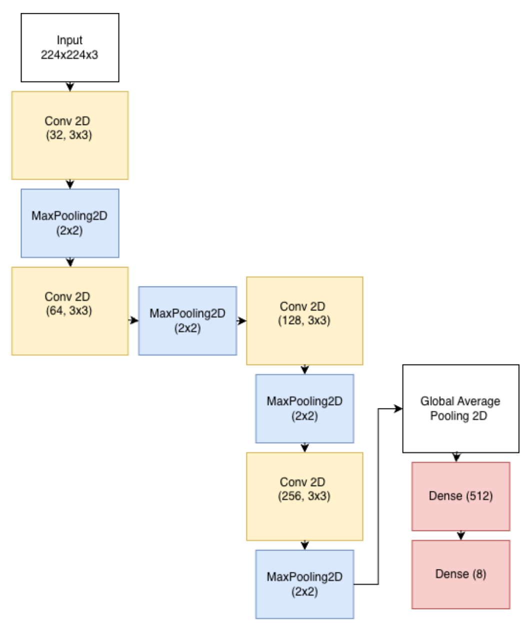

Figure 2. Architecture diagram of the GAP-Regularized Stronger CNN (GAP-SCNN).

Table 3
Training configuration for GAP-SCNN.

Hyperparameter	Value
Loss function	Sparse Categorical Crossentropy
Optimizer	Adam
Initial learning rate	1×10^{-4}
Batch size	64
Maximum epochs	40
Early stopping	Patience = 6
Regularization	Dropout = 0.5, L2 ($\lambda = 10^{-4}$)

Table 4
Model parameter statistics for GAP-SCNN.

Parameter	Value
Total parameters	1,576,200
Trainable parameters	1,574,664
Non-trainable parameters	1,536 (BatchNorm)
Parameter reduction vs. baseline	≈59.2%

5.1.3 Training Configuration

5.1.4 Results Figure 3 shows the training and validation accuracy and loss curves. Early stopping was triggered at epoch 22, with the best validation performance achieved at epoch 18.

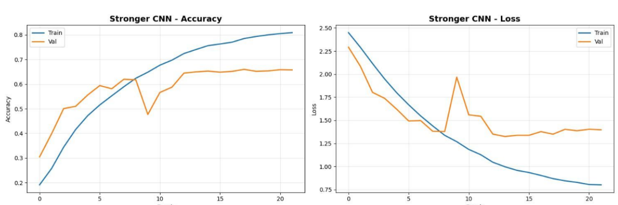

Figure 3. Training and validation accuracy and loss curves for GAP-SCNN.

The final test accuracy achieved by GAP-SCNN was **71.09%**. Class-wise analysis revealed strong performance on *Happy*, *Neutral*, and *Surprise* classes, while lower performance was observed for minority classes such as *Contempt* and *Disgust*, primarily due to class imbalance.

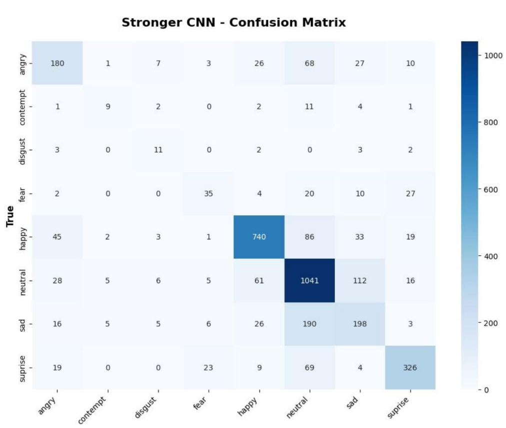

Figure 4. Confusion matrix for GAP-SCNN on the FERPlus test set.

5.2 Baseline CNN (48×48 Grayscale) Ahmed

5.2.1 Data Preprocessing This baseline model follows a native FERPlus-style preprocessing pipeline using low-resolution grayscale images. All preprocessing steps were applied consistently across training, validation, and test splits, except for data augmentation, which was restricted to the training set.

Table 5
Preprocessing pipeline for Baseline CNN.

Step	Details	Rationale
Image size	$48 \times 48 \times 1$	Native FERPlus resolution
Color format	Grayscale	Preserve original dataset format
Normalization	$[0, 255] \to [0, 1]$	Stable optimization
Sampling	Class-aware	Mitigate class imbalance
Batch size	64	Efficient training

Table 6
Training-set data augmentation strategy for Baseline CNN.

Technique	Parameters	Purpose
Horizontal flip	$p = 0.5$	Facial symmetry invariance
Rotation	$\pm 20^\circ$	Robustness to pose variation
Zoom	±10%	Scale invariance
Width / height shift	±10%	Translation robustness

Validation and test sets were not augmented to ensure unbiased evaluation.

5.2.2 Model Architecture The Baseline CNN is a lightweight convolutional neural network trained from scratch, designed to establish a performance reference using low-resolution grayscale inputs. The network consists of three convolutional stages with increasing filter depths (32, 64, 128). Each stage contains two 3×3 convolutional layers followed by Batch Normalization and ReLU activation, after which MaxPooling is applied.

Dropout with a rate of 0.3 is used after each pooling operation to reduce overfitting. The classification head consists of two fully connected layers (512 and 256 units), each followed by Batch Normalization and ReLU activation. A Dropout layer with a rate of 0.5 is applied before the output layer. The final layer is an 8-unit softmax corresponding to FERPlus emotion classes. No pretrained weights were used.

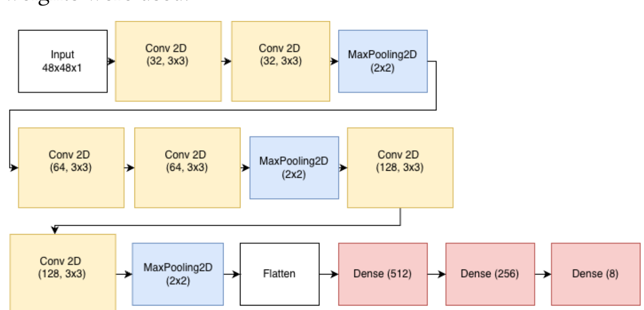

Figure 5. Architecture diagram of the Baseline CNN trained from scratch.

Table 7
Training configuration for Baseline CNN.

Hyperparameter	Value
Loss function	Sparse Categorical Crossentropy
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	64
Maximum epochs	50
Early stopping	Patience = 6
LR reduction	On plateau
Regularization	Dropout, BatchNorm

5.2.3 Training Configuration

5.2.4 Results Figure 6 illustrates the training and validation accuracy and loss curves. Although the model achieved high training accuracy, a noticeable train–validation gap emerged during later epochs, indicating overfitting despite regularization.

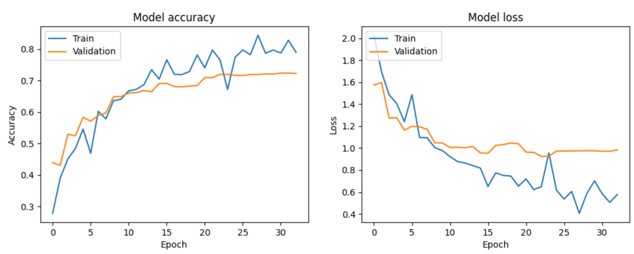

Figure 6. Training and validation accuracy and loss curves for the Baseline CNN.

The final test accuracy achieved by the Baseline CNN was **76.00%**. The training accuracy reached **89.06%**, while validation accuracy plateaued at **67.74%**, resulting in a train–validation gap of **21.32%**. This behavior highlights the limitations of shallow architectures trained from scratch on FERPlus.

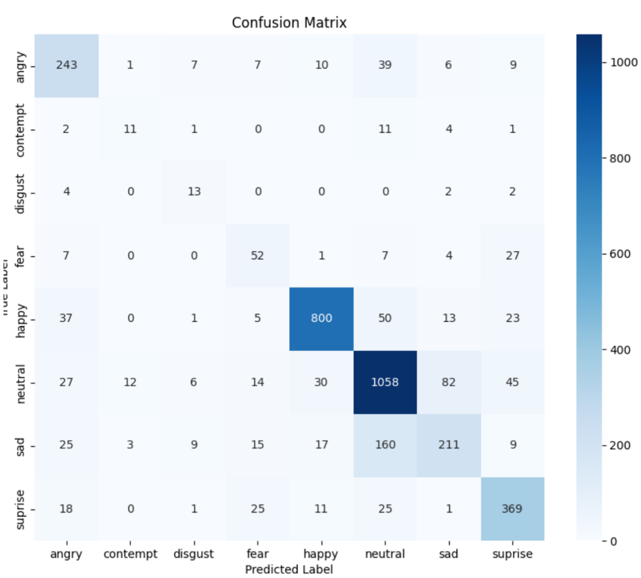

Figure 7. Confusion matrix for the Baseline CNN on the FERPlus test set.

5.3 *Custom Inception-Based CNN (Nada)*

5.3.1 Data Preprocessing The FERPlus dataset was used in its native grayscale format to preserve fine-grained facial details. Images were resized to a fixed resolution and normalized prior to training. Data augmentation was applied exclusively to the training set to reduce overfitting, while validation and test sets remained unaugmented.

Table 8
Preprocessing pipeline for Nada's Inception-based CNN.

Step	Details
Input resolution	$48 \times 48 \times 1$ (grayscale)
Normalization	Pixel values scaled to $[0, 1]$
Data augmentation	Training set only
Batch size	64

Table 9
Training-set data augmentation strategy for Nada's Inception-based CNN.

Technique	Parameters	Purpose
Random rotation	$\pm 30^\circ$	Robustness to pose variation
Random zoom	$\pm 20\%$	Scale invariance
Shear transformation	$\pm 10\%$	Geometric variation
Horizontal flip	Enabled	Facial symmetry invariance
Brightness / contrast	Random	Illumination robustness

5.3.2 Model Architecture This model is a custom Inception-based convolutional neural network trained entirely from scratch. The architecture is designed to capture multi-scale facial features by employing parallel convolutional branches within each Inception module.

The network consists of an initial convolutional stem followed by three stacked Inception modules. Each module includes parallel 1×1, 3×3, and 5×5 convolutional paths, as well as a max-pooling branch. Feature maps from all branches are concatenated and passed to subsequent layers. MaxPooling layers are applied between Inception blocks to reduce spatial dimensions.

A Global Average Pooling layer replaces flattening to reduce parameter count and mitigate overfitting. The classification head consists of two fully connected layers with Batch Normalization, ReLU activation, and Dropout, followed by a softmax output layer corresponding to the eight FERPlus emotion classes.

Table 10
Training configuration for Nada's Inception-based CNN.

Hyperparameter	Value
Optimizer	Adam
Initial learning rate	1×10^{-3}
Learning rate scheduling	ReduceLROnPlateau
Batch size	64
Maximum epochs	100
Early stopping	Patience = 10
Loss function	Categorical Crossentropy + Label Smoothing (0.1)
Regularization	Dropout, L2 ($\lambda = 5 \times 10^{-3}$), BatchNorm

5.3.3 Training Configuration

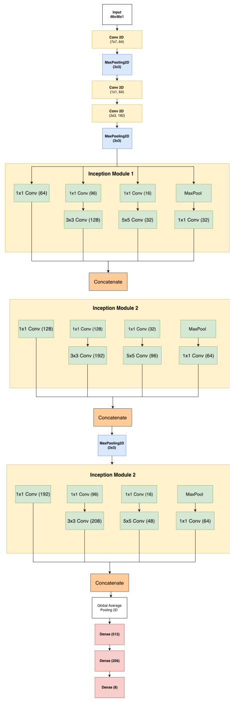

Figure 8. Architecture diagram of Nada's custom Inception-based CNN.

5.3.4 Results The model achieved a final test accuracy of **72.21%**. Training and validation curves indicate rapid convergence during early epochs, followed by stable learning with a reduced training–validation gap due to strong regularization.

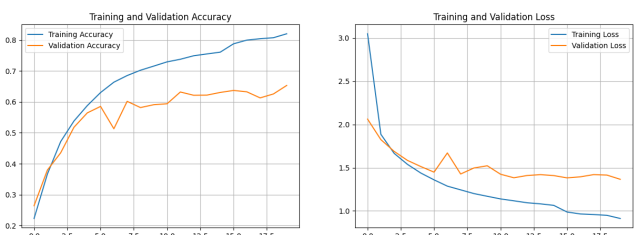

Figure 9. Training and validation accuracy and loss curves for Nada's Inception-based CNN.

Class-wise performance analysis shows strong recognition of highly expressive emotions such as *Happy* and *Neutral*, while confusion persists between visually similar classes including *Fear* and *Surprise*, and *Neutral* and *Sad*.

Interpretability. Visualization techniques such as Grad-CAM were used to analyze model attention. Correct predictions exhibit focus on semantically meaningful facial regions including the eyes and mouth, while misclassified samples reveal overlapping attention patterns between subtle emotions.

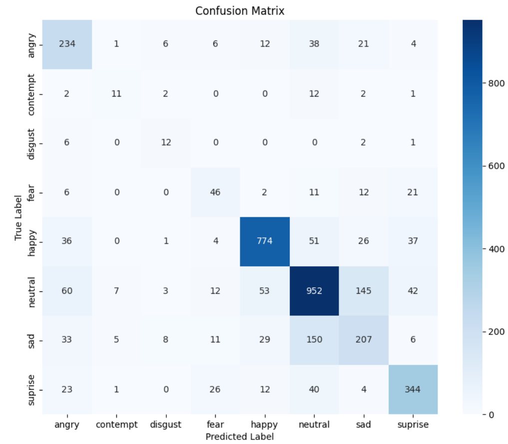

Figure 10. Confusion matrix for Nada's Inception-based CNN on the FERPlus test set.

Figure 11. Grad-CAM visualizations for correct and misclassified samples.

5.4 *Custom Inception–ResNet CNN (Aly Zaki)*

5.4.1 Data Preprocessing The FERPlus dataset was processed using its native grayscale format to preserve fine-grained facial details. Prior to training, extensive dataset cleaning was performed to remove duplicate samples and prevent cross-split leakage. The model was trained on a reduced 7-class setup after excluding the *Contempt* class due to severe class imbalance.

Table 11
Preprocessing pipeline for Aly Zaki's Inception–ResNet CNN.

Step	Details
Input resolution	$48 \times 48 \times 1$ (grayscale)
Normalization	Pixel values scaled to $[0, 1]$
Class setup	7 emotion classes (Contempt removed)
Dataset cleaning	Duplicate and leakage removal
Batch size	64

Table 12
Training-set data augmentation strategy for Aly Zaki's model.

Technique	Parameters	Purpose
Random rotation	$\pm 15^\circ$	Handle pose variation
Width / height shift	$\pm 10\%$	Translation robustness
Random zoom	$\pm 10\%$	Scale invariance
Horizontal flip	Enabled	Facial symmetry

5.4.2 Model Architecture This model is a custom Inception–ResNet-inspired convolutional neural network trained entirely from scratch. The architecture combines multi-scale feature extraction through Inception-style branches with residual connections to improve gradient flow and training stability.

The network consists of an initial convolutional stem followed by multiple Inception–ResNet blocks with increasing representational capacity. SpatialDropout is applied progressively to mitigate overfitting. A Global Average Pooling layer replaces flattening to reduce parameter count and improve generalization. The classification head includes Batch Normalization, Dropout, and a softmax output layer corresponding to the seven emotion classes.

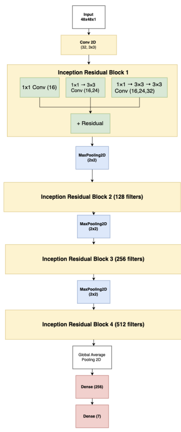

Figure 12. Architecture diagram of Aly Zaki's custom Inception–ResNet CNN.

Table 13
Training configuration for Aly Zaki's Inception–ResNet CNN.

Hyperparameter	Value
Loss function	Categorical Crossentropy + Label Smoothing (0.1)
Optimizer	Adam
Initial learning rate	1×10^{-3}
Learning rate schedule	Cosine annealing
Batch size	64
Maximum epochs	100
Early stopping	Patience = 15
Regularization	L2 ($\lambda = 5 \times 10^{-4}$), Dropout

5.4.3 Training Configuration

5.4.4 Results Figure 13 shows the training and validation accuracy and loss curves. The curves closely follow each other throughout training, indicating stable optimization and effective regularization.

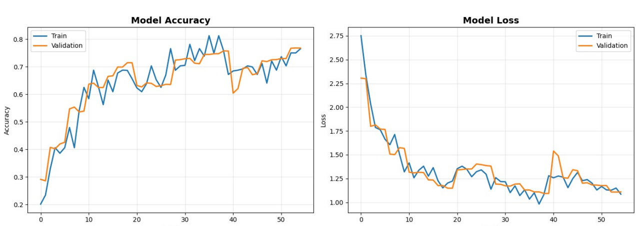

Figure 13. Training and validation accuracy and loss curves for Aly Zaki's Inception–ResNet CNN.

The model achieved a final test accuracy of **76.29%**. Performance was strongest for highly expressive emotions such as *Happy* and *Surprise*, while confusion persisted between visually similar classes such as *Sad* and *Neutral*.

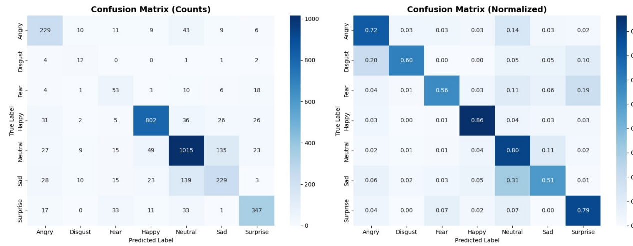

Figure 14. Confusion matrix for Aly Zaki's Inception–ResNet CNN on the FERPlus test set.

5.4.5 Model Interpretability Grad-CAM visualizations were used to analyze model attention. Correct predictions show consistent focus on semantically relevant facial regions such as the eyes, eyebrows, and mouth.

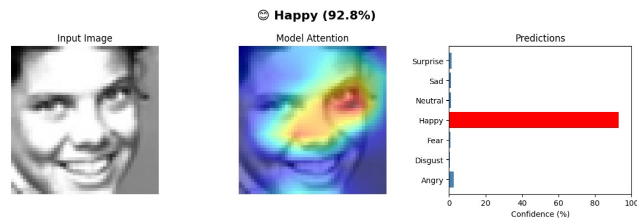

Figure 15. Grad-CAM visualizations for correct predictions across emotion classes.

Misclassified samples reveal overlapping attention regions between subtle emotions, highlighting intrinsic ambiguity within the FERPlus dataset.

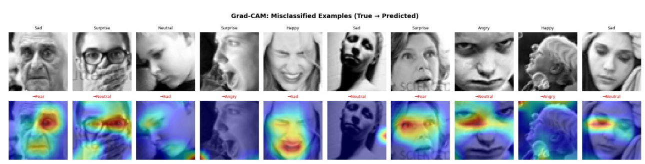

Figure 16. Grad-CAM visualizations for misclassified emotion samples.

6 Methodology (Transfer Learning))

6.1 ResNet50 Feature Extraction Model (Phase 1)

6.1.1 Data Preprocessing The FERPlus dataset was used in its original grayscale format and adapted to meet the input requirements of ResNet50. All preprocessing steps were applied consistently across training, validation, and test sets, with no data augmentation used in this phase to ensure stable feature extraction.

Table 14
Preprocessing pipeline for ResNet50 feature extraction model.

Step	Description
Input resolution	48×48 grayscale FERPlus images
Color conversion	Grayscale $\longrightarrow$ RGB (3 channels)
Image resizing	$48 \times 48 \longrightarrow 224 \times 224$
Normalization	ResNet50 `preprocess_input`
Data augmentation	None (Phase 1 strategy)
Class balancing	Not required (balanced dataset)

Figure 17 illustrates the class distribution of the FERPlus training set, showing a nearly balanced dataset across the eight emotion categories, with a slightly higher proportion of neutral samples.

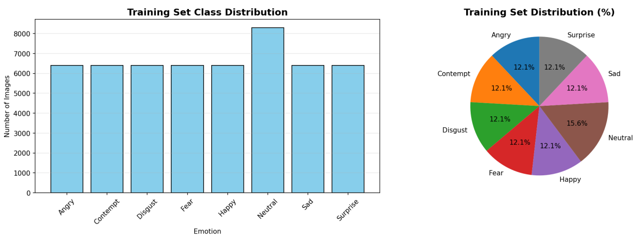

Figure 17. Training set class distribution for the FERPlus dataset.

6.1.2 Model Architecture This model follows a transfer learning approach using ResNet50 pretrained on ImageNet as a fixed feature extractor. All convolutional layers were frozen during training to prevent overfitting and to leverage robust low- and mid-level visual representations.

The classification head consists of a Global Average Pooling layer followed by fully connected layers tailored to FERPlus emotion classification.

Table 15
Architecture configuration of the ResNet50 feature extraction model.

Component	Configuration
Backbone	ResNet50 (ImageNet pre-trained)
Frozen layers	100% of convolutional layers
Pooling	Global Average Pooling
Dense layer	512 units, ReLU
Dropout	0.5
Output layer	Dense (8), Softmax
Input shape	$224 \times 224 \times 3$

Table 16
Training configuration for ResNet50 feature extraction model.

Hyperparameter	Value
Optimizer	Adam
Learning rate	1×10^{-3}
Batch size	32
Epochs	30
Loss function	Categorical Crossentropy
Backbone training	Frozen
Callbacks	EarlyStopping, ReduceLROnPlateau, ModelCheckpoint, CSVLogger

Table 17
Model parameter statistics for the ResNet50 feature extraction model.

Parameter	Value
Total parameters	24,115,336
Trainable parameters	527,112
Non-trainable parameters	23,588,224
Dropout rate	0.5

6.1.3 Training Configuration

6.1.4 Results Figure 18 presents the training and validation accuracy and loss curves. The model exhibits stable convergence with a very small training–validation gap, indicating strong generalization and no overfitting.

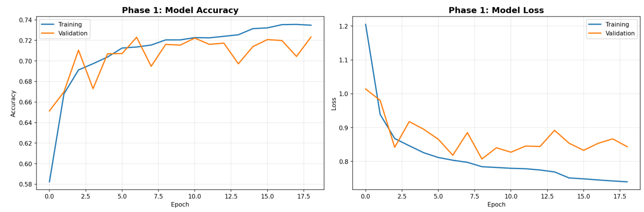

Figure 18. Training and validation accuracy and loss curves for the ResNet50 feature extraction model.

The best validation accuracy achieved was **72.33%**, while the final test accuracy reached **69.41%**. These results are considered strong for a frozen-backbone Phase 1 model trained without data augmentation.

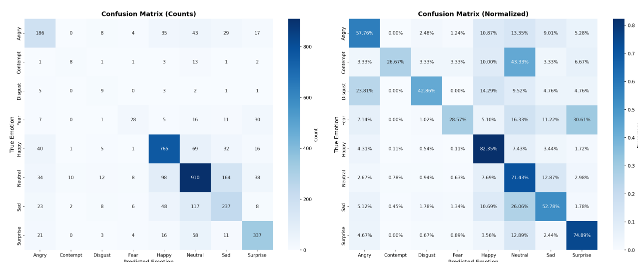

Figure 19. Confusion matrix for the ResNet50 feature extraction model on the FERPlus test set.

Per-class performance metrics are shown in Figure 20. The model performs best on *Happy*, *Neutral*, and *Surprise* classes, while lower scores are observed for *Contempt*, *Disgust*, and *Fear*, which are known to be challenging in FER tasks.

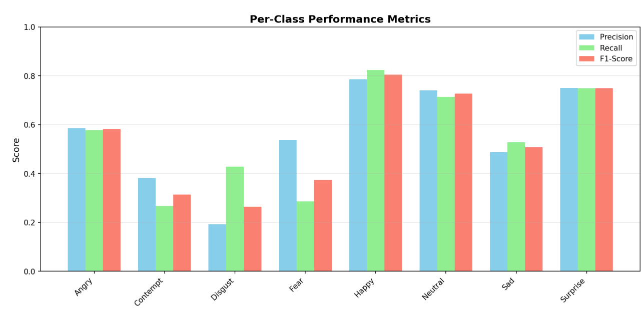

Figure 20. Per-class precision, recall, and F1-score for the ResNet50 feature extraction model.

Figure 21 presents representative misclassified examples. Common confusion patterns include *Fear* versus *Surprise*, *Neutral* versus *Sad*, and *Disgust* versus *Angry*, consistent with known FERPlus and FER2013 dataset challenges.

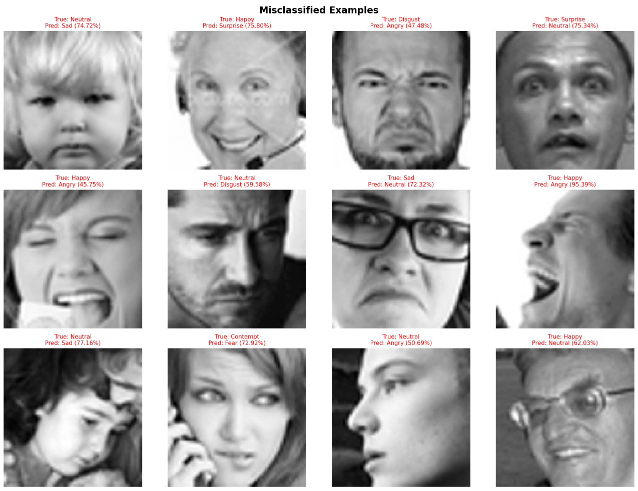

Figure 21. Examples of misclassified facial expressions from the FERPlus test set.

6.2 EfficientNetB3 Feature Extraction Model (Phase 1)

6.2.1 Data Preprocessing FERPlus grayscale images were converted to three-channel RGB format to satisfy the input requirements of EfficientNetB3. Images were resized to the network's expected resolution and normalized prior to training. Data augmentation was applied exclusively to the training set, while validation and test sets remained unaugmented to ensure unbiased evaluation.

Table 18
Preprocessing pipeline for EfficientNetB3 (Phase 1).

Step	Details
Input image size	$224 \times 224 \times 3$
Color format	RGB (converted from grayscale)
Normalization	$[0, 255] \rightarrow [0, 1]$
Data augmentation	Applied to training set only
Class balancing	Not applied

Table 19
Training-set data augmentation strategy for EfficientNetB3.

Technique	Parameters	Purpose
Random horizontal flip	$p = 0.5$	Facial symmetry invariance
Random rotation	$\pm 10^\circ$	Robustness to pose variation
Random zoom	$\pm 10\%$	Scale robustness

6.2.2 Model Architecture This model employs EfficientNetB3 pretrained on ImageNet as a fixed feature extractor. EfficientNet utilizes compound scaling to balance network depth, width, and input resolution, enabling strong representational capacity with improved parameter efficiency.

During Phase 1 training, the entire EfficientNetB3 backbone was frozen to preserve pretrained representations and prevent overfitting. A Global Average Pooling layer was applied to the backbone output, followed by a task-specific classification head composed of a fully connected layer with Batch Normalization, ReLU activation, and Dropout. The final output layer uses a softmax activation corresponding to the eight FERPlus emotion classes.

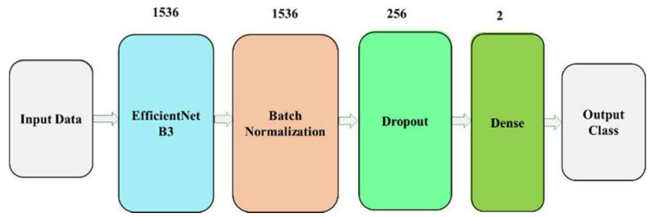

Figure 22. Architecture diagram of the EfficientNetB3 feature extraction model.

Table 20
Training configuration for EfficientNetB3 (Phase 1).

Hyperparameter	Value
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	32
Epochs	20
Loss function	Sparse Categorical Crossentropy
Backbone training	Fully frozen
Regularization	Dropout (0.5), Batch Normalization

6.2.3 Training Configuration

6.2.4 Results Figure 23 presents the training and validation accuracy and loss curves for EfficientNetB3 Phase 1. The model demonstrates stable convergence, with validation accuracy improving steadily before reaching a plateau. The moderate gap between training and validation curves indicates reasonable generalization under a frozen-backbone strategy.

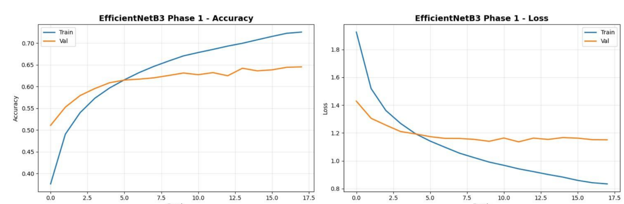

Figure 23. Training and validation accuracy and loss curves for EfficientNetB3 Phase 1.

Fine-Tuning Observation. A second training phase involving partial fine-tuning of the EfficientNetB3 backbone was explored. While training accuracy increased significantly, validation performance degraded and validation loss increased, indicating pronounced overfitting. For this reason, only Phase 1 results are considered reliable and are reported in this study. Visualization of Phase 2 behavior is provided for qualitative comparison only. The fine-tuning behavior is visualized in Figure 24.

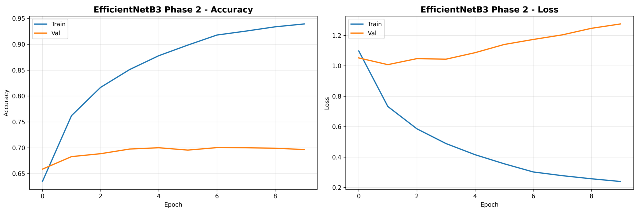

Figure 24. Training and validation accuracy and loss curves for EfficientNetB3 Phase 2 (partial fine-tuning). A widening train–validation gap and increasing validation loss indicate severe overfitting.

Table 21
Preprocessing pipeline for DenseNet121 transfer learning.

Step	Details
Original image size	48 × 48 × 1 (grayscale)
Color conversion	Grayscale ⟶ RGB (pseudo-RGB)
Image resizing	48 × 48 ⟶ 96 × 96
Normalization	ImageNet mean / std normalization
Class filtering	Contempt removed (7-class setup)
Data cleaning	Duplicate and leakage removal (pHash-based)
Batch size	32

Table 22
Training-set data augmentation for DenseNet121.

Technique	Parameters	Purpose
Random rotation	±15°	Pose robustness
Width / height shift	±10%	Translation invariance
Random zoom	±10%	Scale robustness
Horizontal flip	Enabled	Facial symmetry

6.3 DenseNet121 Transfer Learning Model (Phase 1 + Phase 2)

6.3.1 Motivation This experiment evaluates the effectiveness of DenseNet121-based transfer learning for facial emotion recognition on the FERPlus dataset. DenseNet121 was selected due to its dense connectivity pattern, which encourages feature reuse, improves gradient flow, and enhances parameter efficiency. The goal was to assess whether ImageNet-pretrained representations can generalize effectively to FER under a controlled fine-tuning strategy.

6.3.2 Data Preprocessing FERPlus grayscale images were adapted to meet the input requirements of DenseNet121. Since the backbone was pretrained on RGB ImageNet images, grayscale inputs were converted into pseudo-RGB format by channel replication. Images were resized to satisfy the minimum spatial resolution required by the network.

Data augmentation was applied exclusively to the training set.

6.3.3 Dataset Cleaning Perceptual hashing (pHash) was used to detect and remove duplicate samples, cross-split leakage, and label inconsistencies. Approximately **6,931 images** (10% of the dataset) were removed, resulting in improved training stability and more reliable evaluation.

6.3.4 Model Architecture The model follows a transfer learning paradigm using DenseNet121 pretrained on ImageNet as the backbone. A compact classification head was attached to adapt the network for FER:

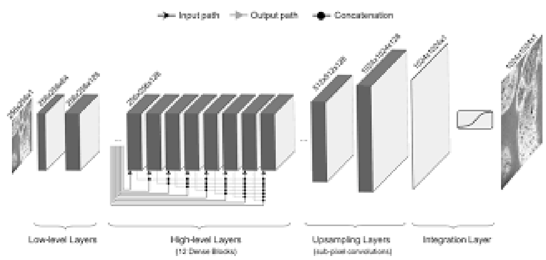

Figure 25. DenseNet121 architecture with the custom classification head.

Table 23
Two-phase training configuration for DenseNet121.

Parameter	Phase 1	Phase 2
Optimizer	Adam	Adam
Learning rate	1×10^{-3}	1×10^{-5}
LR scheduling	Cosine Annealing	ReduceLROnPlateau
Epochs	19	30
Loss function	Cat. CE + Label Smoothing (0.1)	Same
Backbone	Frozen	15% unfrozen

Table 24
Model parameter statistics for DenseNet121.

Component	Parameters
DenseNet121 backbone	7,037,504
Classification head	661,127
Total parameters	7,698,503
Trainable (Phase 2)	2,006,471

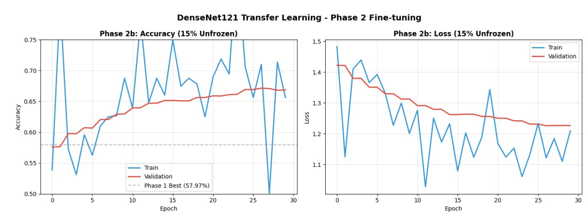

Figure 26. Training and validation accuracy and loss curves for DenseNet121.

- Global Average Pooling
- Dense (512 units) + BatchNorm + ReLU + Dropout (0.5)
- Dense (256 units) + BatchNorm + ReLU + Dropout (0.3)
- Dense (7 units) with Softmax activation

6.3.5 Training Strategy A two-phase training strategy was employed:

- **Phase 1:** Backbone frozen; only the classification head trained.
- **Phase 2:** Top 15% of DenseNet121 layers unfrozen for controlled fine-tuning.

6.3.6 Model Parameters

6.3.7 Results The DenseNet121 model achieved a final test accuracy of **67.83%**. Controlled fine-tuning in Phase 2 led to gradual validation improvements while maintaining a modest train–validation gap of approximately 4%, indicating stable generalization.

6.3.8 Discussion DenseNet121 benefits from pretrained ImageNet features and dense connectivity, yielding stable generalization behavior. However, its performance remains slightly below the strongest from-scratch architectures, suggesting that domain mismatch limits peak accuracy. Partial fine-tuning proved effective, whereas more aggressive unfreezing would likely result in overfitting.

6.4 MobileNetV2 Feature Extraction Model (Phase 1)

6.4.1 Data Preprocessing The FERPlus dataset was processed to satisfy the input requirements of MobileNetV2. Original grayscale facial images were converted to three-channel RGB format and resized to the network's expected resolution. Data augmentation was applied exclusively to the training set, while validation and test sets were left unaugmented to ensure unbiased evaluation.

Table 25
Preprocessing pipeline for MobileNetV2 feature extraction model.

Step	Details
Original image size	$48 \times 48 \times 1$ (grayscale)
Color conversion	Grayscale $\longrightarrow$ RGB (3 channels)
Image resizing	$48 \times 48 \longrightarrow 224 \times 224$
Normalization	MobileNetV2 preprocessing ($[-1, 1]$)
Data augmentation	Training set only
Batch size	32

Table 26
Training-set data augmentation strategy for MobileNetV2.

Technique	Parameters	Purpose
Random horizontal flip	$p = 0.5$	Facial symmetry invariance
Random rotation	$\pm 20^\circ$	Robustness to pose variation
Random zoom	$\pm 10\%$	Scale robustness
Width / height shift	$\pm 10\%$	Translation robustness

6.4.2 Model Architecture This model adopts a transfer learning approach using MobileNetV2 pretrained on ImageNet as a fixed feature extractor. MobileNetV2 employs depthwise separable convolutions and inverted residual blocks, enabling efficient feature learning with reduced computational cost.

During Phase 1 training, the entire MobileNetV2 backbone was frozen to preserve pretrained representations and limit overfitting. A Global Average Pooling layer was applied to the backbone output, followed by a task-specific classification head composed of fully connected layers with Batch Normalization and Dropout. The final output layer uses a softmax activation corresponding to the eight FERPlus emotion classes.

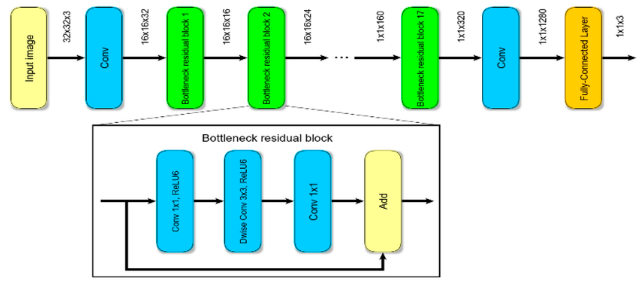

Figure 27. Architecture of the MobileNetV2 network illustrating inverted residual bottleneck blocks with depthwise separable convolutions.

Table 27
Training configuration for MobileNetV2 (Phase 1).

Hyperparameter	Value
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	32
Epochs	20
Loss function	Sparse Categorical Crossentropy
Backbone training	Fully frozen
Regularization	Dropout (0.5), Batch Normalization

Table 28
Model parameter statistics for MobileNetV2 feature extraction model.

Parameter	Value
Total parameters	3,538,984
Trainable parameters	263,432
Non-trainable parameters	3,275,552
Trainable ratio	≈7.4%

6.4.3 Training Configuration

6.4.4 Results The MobileNetV2 feature extraction model demonstrated stable convergence throughout training, with a small gap between training and validation accuracy. The frozen-backbone strategy resulted in reliable generalization but limited adaptation to emotion-specific facial features.

The final test accuracy achieved by the MobileNetV2 Phase 1 model was approximately **68%**. Performance was strongest for expressive emotions such as *Happy* and *Surprise*, while reduced accuracy was observed for subtle or minority classes, consistent with known FERPlus challenges.

Visualization of training curves, confusion matrices, and class-wise performance metrics are provided separately.

7 Comprehensive Comparative Analysis and Conclusions

This section presents a unified comparative analysis of all evaluated deep learning models on the FERPlus dataset. The analysis is structured around learning strategy, generalization behavior, accuracy ceilings, and computational efficiency. To improve clarity and interpretability, results are summarized using complementary full-width tables rather than extended prose discussion.

7.1 Overview of Learning Strategies

The evaluated models naturally fall into three learning paradigms: models trained from scratch using FERPlus as the sole source of supervision, transfer learning models using frozen ImageNet-pretrained backbones, and partially fine-tuned transfer learning models that allow limited domain adaptation. Each paradigm exhibits distinct behavior in terms of stability, adaptability, and performance ceiling.

Table 29
Categorization of evaluated models by learning strategy.

Category	Training Strategy	Key Characteristics
From Scratch	Full supervision on FERPlus	High adaptability, higher overfitting risk
Transfer Learning	Frozen ImageNet backbone	Strong stability, limited task adaptation
Partial Fine-Tuning	Controlled unfreezing	Balanced adaptation and generalization

7.2 Performance Comparison: Models Trained From Scratch

Models trained from scratch exhibit the widest performance variability, reflecting their strong dependence on architectural design and regularization. Shallow baseline models achieve high training accuracy but suffer from severe overfitting, while architectures incorporating structural regularization mechanisms demonstrate improved stability.

Table 30
Comparison of models trained from scratch on FERPlus.

Model	Input	Classes	Params (M)	Test Acc.	Gen. Gap	Stability
Baseline CNN (Ahmed)	48×48	8	∼2.5	76.00%	High	Low
GAP-SCNN (Bayoumi)	224×224	8	1.58	71.09%	Low	High
Inception-CNN (Nada)	48×48	8	∼3.1	72.21%	Moderate	Medium
Inception–ResNet CNN (Aly)	48×48	7	∼4.2	**76.29%**	Low	High

Among all from-scratch models, the Inception–ResNet CNN achieves the strongest balance between accuracy and generalization. The combination of multi-scale feature extraction and residual connections enables efficient representation learning while maintaining smooth gradient flow, resulting in the highest test accuracy with a consistently small generalization gap.

7.3 Performance Comparison: Transfer Learning Models

Transfer learning models consistently demonstrate superior training stability and minimal divergence between training and validation curves. However, they exhibit a clear performance ceiling compared to the strongest from-scratch architectures, indicating limited task-specific adaptation under frozen-backbone settings.

Table 31
Comparison of transfer learning models on FERPlus.

Model	Input	Classes	Params (M)	Test Acc.	Gen. Gap	Efficiency
ResNet50 (Nada)	224×224	8	24.1	69.41%	Very Low	Medium
EfficientNetB3 (Bayoumi)	224×224	8	12.2	∼70%	Low	High
DenseNet121 (Aly)	96×96	7	7.7	67.83%	Low	Medium
MobileNetV2 (Ahmed)	224×224	8	3.5	∼68%	Very Low	**Very High**

Frozen-backbone transfer learning yields robust generalization but restricts adaptation to fine-grained facial emotion cues. DenseNet121 benefits from feature reuse and connectivity, yet remains constrained by domain mismatch. EfficientNetB3 and MobileNetV2 highlight the effectiveness of parameter-efficient designs for deployment under computational constraints.

7.4 Cross-Model Behavioral Trends

Despite architectural diversity, consistent behavioral patterns emerge across all models, suggesting intrinsic dataset-level challenges rather than architecture-specific limitations.

Table 32
Observed cross-model behavioral patterns on FERPlus.

Observation	Interpretation
Fear–Surprise confusion	Intrinsic dataset ambiguity
Neutral–Sad confusion	Subtle expression overlap
Reduced-class configurations (7-class)	Improved training stability
Aggressive fine-tuning	Increased overfitting risk

7.5 Accuracy–Stability–Efficiency Trade-Off

A clear trade-off exists between peak accuracy, generalization stability, and computational efficiency. No single model dominates across all dimensions.

Table 33
Accuracy–stability–efficiency trade-off summary.

Model Category	Accuracy	Stability	Efficiency
Baseline CNNs	High	Low	Medium
Regularized From-Scratch	High	High	Medium
Frozen Transfer Learning	Medium	Very High	Medium
Lightweight Transfer Learning	Medium	High	Very High

7.6 Final Conclusions

This comparative study demonstrates that no single architecture dominates across all evaluation dimensions. Carefully regularized from-scratch models achieve the highest accuracy, with the Inception–ResNet CNN providing the best overall balance between performance and stability. Transfer learning models offer reliable generalization and reduced sensitivity to hyperparameter tuning, making them well-suited for resource-constrained or deployment-oriented scenarios. Lightweight transfer learning models such as MobileNetV2 provide an attractive compromise between efficiency and robustness.

Acknowledgements

This work was conducted as part of the Deep Learning course offered by the Faculty of Computing and Information Systems at Egypt University of Informatics. The authors would like to express their sincere gratitude to Dr. Mona Soliman for her guidance, support, and valuable feedback throughout the development of this project.